UNIVERSITY PHYSICS MASTER SERIES • VOLUME 1

Kinematics & Two-Dimensional Motion

Vector Algebra & Coordinate Calculus • 1D Kinematic Equations & SUVAT Derivations • Projectile Motion Dynamics • Range, Apex, & Complementary Symmetry • Uniform & Non-Uniform Circular Kinematics • Relative Motion & Galilean Reference Frames

---

Synthesized From:

• University Lecture Series — *Chapter 1.2: 2D Motion & Circular Kinematics*
• R. Shankar — *Fundamentals of Physics: Mechanics, Relativity, and Thermodynamics* (Chapters 1 & 2)
• University Midterm Physics Reference Standards & Vector Mechanics Formulations

Publication Standard: Pure English Master Edition • Document/Physics/ Directory
Includes: 4 Scalable Vector SVG Diagrams • 6 Hand-Calculation Fast-Tracks • 5 Rigorous Worked Problems • SI Verification

Table of Contents

Part I: Vector Foundations & One-Dimensional Kinematics

Chapter 1: Vectors, Coordinate Bases, and Vector Calculus VECTORS

Chapter 2: Kinematic Definitions: Position, Velocity, and Acceleration

Chapter 3: Motion with Constant Acceleration & Calculus SUVAT Derivations TRICKS K1, K2

Part II: Two-Dimensional Kinematics & Projectile Motion

Chapter 4: Orthogonal Motion Independence & 2D Vector Equations

Chapter 5: Projectile Motion Analytical Theory: Time, Apex, and Range FIG 1.1 • TRICK K3

Chapter 6: Trajectory Parabola, Launch Angle Symmetry, & Asymmetric Elevations FIG 1.4 • TRICK K4

Part III: Uniform and Non-Uniform Circular Kinematics

Chapter 7: Circular Coordinate Variables, Angular Speed, and Frequency

Chapter 8: Centripetal (Radial) Acceleration Geometric & Vector Proofs

Chapter 9: Non-Uniform Circular Kinematics, Tangential Acceleration, & Curvature FIG 1.2 • TRICK K5

Part IV: Relative Motion & Reference Frames

Chapter 10: Inertial Reference Frames & Galilean Position Transformations

Chapter 11: Galilean Velocity Addition: River Crossing & Aerial Crosswind Vectors FIG 1.3 • TRICK K6

Part V: Step-by-Step Worked Problems with Fast-Tracks

Problem 1: Level-Ground Projectile Motion with Full Velocity Decomposition FOOTBALL KICK

Problem 2: Cliff Launch with Asymmetric Landing Elevation SUPPLY DROP

Problem 3: Arthur Ashe Stadium Tennis Hit into Elevated Grandstand QUADRATIC TIME

Problem 4: Non-Uniform Circular Acceleration of a Supercar on a Curved Track RADIAL + TANGENT

Problem 5: 2D Relative Velocity in Crosswind Aerial Navigation VECTOR HEADING

Part VI: Master Reference Summary & Formula Sheet

Chapter 12: Comprehensive Formula Matrix, Fast-Track Summary, & Exam Traps FORMULAS

Part I: Vector Foundations & One-Dimensional Kinematics

Chapter 1: Vectors, Coordinate Bases, and Vector Calculus

Classical Newtonian mechanics begins with a precise mathematical vocabulary for describing where bodies are located and how their spatial configurations evolve across time. Physical quantities are fundamentally divided into two categories: **scalars**, which are completely characterized by a numerical magnitude and physical units (such as mass m, time t, temperature T, and energy E), and **vectors**, which possess both a magnitude and a spatial direction, transforming under coordinate rotations according to definite geometrical rules (such as position $\mathbf{r}$, displacement $\Delta\mathbf{r}$, velocity $\mathbf{v}$, acceleration $\mathbf{a}$, and force $\mathbf{F}$).

1.1 Cartesian Basis Vectors and Component Representation

In a three-dimensional Euclidean space $\mathbb{R}^3$ equipped with a right-handed Cartesian coordinate system, we define three orthogonal unit basis vectors: $\hat{\mathbf{i}}$ (pointing along the positive x-axis), $\hat{\mathbf{j}}$ (along positive y), and $\hat{\mathbf{k}}$ (along positive z). Any arbitrary vector $\mathbf{A}$ can be uniquely expressed as an invariant linear combination of these basis elements:

$$\mathbf{A} = A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}$$

The scalar components A_x, A_y, and A_z represent the orthogonal projections of $\mathbf{A}$ onto the respective coordinate axes. The scalar magnitude (Euclidean norm) $|\mathbf{A}|$ is defined by Pythagoras's theorem:

$$|\mathbf{A}| = A = \sqrt{(A_x{}^2 + A_y{}^2 + A_z{}^2)}$$

For a planar vector constrained to the xy-plane ($d = 2$), the components are related to the magnitude A and counterclockwise direction angle θ measured from the positive x-axis by:

$$A_x = A\cos\theta, \quad A_y = A\sin\theta, \quad \theta = \arctan\frac{A_y}{A_x}$$

1.2 Vector Products: Scalar (Dot) and Vector (Cross) Products

Two fundamental multiplicative operations exist between vectors:

1. The Scalar (Dot) Product: The dot product produces a scalar measure of directional alignment and projection:

$$\mathbf{A}\cdot\mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\varphi = A_xB_x + A_yB_y + A_zB_z$$

where φ is the angle between $\mathbf{A}$ and $\mathbf{B}$ ($0 \le \varphi \le \pi$). Crucially, $\mathbf{A}\cdot\mathbf{B} = 0$ if and only if non-zero vectors $\mathbf{A}$ and $\mathbf{B}$ are strictly orthogonal (perpendicular).

2. The Vector (Cross) Product: The cross product yields a vector strictly perpendicular to both operands, satisfying the right-hand rule with magnitude equal to the area of the spanned parallelogram:

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin\varphi\, \hat{\mathbf{n}} = (A_y B_z - A_z B_y)\hat{\mathbf{i}} + (A_z B_x - A_x B_z)\hat{\mathbf{j}} + (A_x B_y - A_y B_x)\hat{\mathbf{k}}$$

1.3 Vector Calculus: Rates of Change of Vector Functions

When a particle moves through space, its position vector is a vector-valued function of continuous time: $\mathbf{r}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}}$. Because the Cartesian unit vectors $\{\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}\}$ maintain fixed orientations and unit lengths in an inertial frame, their time derivatives vanish identically: $d\hat{\mathbf{i}}/dt = d\hat{\mathbf{j}}/dt = d\hat{\mathbf{k}}/dt = \mathbf{0}$. Consequently, differentiation and integration distribute directly across Cartesian components:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\hat{\mathbf{i}} + \frac{dy}{dt}\hat{\mathbf{j}} + \frac{dz}{dt}\hat{\mathbf{k}} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} + v_z\hat{\mathbf{k}}$$

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2} = \frac{dv_x}{dt}\hat{\mathbf{i}} + \frac{dv_y}{dt}\hat{\mathbf{j}} + \frac{dv_z}{dt}\hat{\mathbf{k}} = a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}} + a_z\hat{\mathbf{k}}$$

Chapter 2: Kinematic Definitions: Position, Velocity, and Acceleration

Kinematics describes motion geometrically without reference to the mass of the moving object or the dynamical forces causing the motion. In one dimension along an axis x, we distinguish between path distance and displacement:

- **Position $x(t)$:** The spatial coordinate of the particle relative to an agreed coordinate origin at time t.
- **Displacement Δx:** The net vector change in position over a discrete time interval $\Delta t = t_2 - t_1$:

$$\Delta x = x(t_2) - x(t_1)$$

  Displacement depends solely on endpoints, whereas total distance traveled s is the scalar arc-length integral $\int |v(t)|\, dt \geq |\Delta x|$.
- **Average Velocity v_{avg}:** The ratio of displacement to elapsed time:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}$$

- **Instantaneous Velocity $v(t)$:** The continuous time limit as $\Delta t \to 0$, defined as the local derivative dx/dt. Geometrically, $v(t)$ is the tangent slope of the position-versus-time graph $x(t)$.
- **Average Acceleration a_{avg}:** The rate of change of velocity:

$$a_{\text{avg}} = \frac{\Delta v}{\Delta t} = \frac{v(t_2) - v(t_1)}{t_2 - t_1}$$

- **Instantaneous Acceleration $a(t)$:** The second time derivative $d^2x/dt^2 = dv/dt$, representing the slope of the velocity curve $v(t)$.

Chapter 3: Motion with Constant Acceleration & Calculus SUVAT Derivations

The most common foundational motion in physics occurs when acceleration is invariant with respect to time: $a(t) = a = \text{const}$. Integrating this differential relation provides the celebrated constant-acceleration kinematic equations (SUVAT):

Theorem 1.1: The Big Five Constant Acceleration Kinematic Equations (SUVAT)

For one-dimensional motion with uniform acceleration a, initial position x_0, initial velocity v_0 at $t = 0$, and final velocity v at time t:

#	Kinematic Formula	Independent Missing Variable
Eq. 1	$\boldsymbol{v = v_0 + at}$	Displacement (Δx omitted)
Eq. 2	$\boldsymbol{\Delta x = \frac{v_0 + v}{2} t}$	Acceleration (a omitted)
Eq. 3	$\boldsymbol{\Delta x = v_0 t + \frac{1}{2} a t^2}$	Final Velocity (v omitted)
Eq. 4	$\boldsymbol{\Delta x = vt - \frac{1}{2} a t^2}$	Initial Velocity (v_0 omitted)
Eq. 5	$\boldsymbol{v^2 = v_0^2 + 2a\Delta x}$	Time (t omitted)

3.1 Rigorous Calculus Derivation of the Kinematic Equations

Derivation of Eq. 1: By definition, $a = dv/dt$. Separating variables and integrating from $t = 0$ (where $v = v_0$) to time t:

$$\int_{v_0}^{v} dv' = \int_0^t a\, dt' \quad \Rightarrow \quad v - v_0 = at \quad \Rightarrow \quad \boldsymbol{v(t) = v_0 + at}$$

Derivation of Eq. 3: Substituting $v(t) = dx/dt$ into Eq. 1 and integrating:

$$\int_{x_0}^{x} dx' = \int_0^t (v_0 + at')\, dt' \quad \Rightarrow \quad x(t) - x_0 = v_0 t + \frac{1}{2} a t^2$$

Derivation of Eq. 5 (The Work-Kinematic Chain Rule Identity): Using the chain rule to eliminate explicit time dependence: $a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$. Rearranging and integrating:

$$\int_{x_0}^{x} a\, dx' = \int_{v_0}^{v} v'\, dv' \quad \Rightarrow \quad a(x - x_0) = \frac{1}{2}(v^2 - v_0^2) \quad \Rightarrow \quad \boldsymbol{v^2 = v_0^2 + 2a\Delta x}$$

TRICK K1 The "5-Second Drop" Rule & Mental Free-Fall Math

Mental Math

Standard Slow Method: Setting up $y = \frac{1}{2} g t^2$ with $g = 9.80\ \text{m/s}^2$ and multiplying $4.90 \times t^2$ by hand.

⚡ **Fast Hand Calculation Shortcut:** Approximate $g \approx 10\ \text{m/s}^2$. Then:

$$\textbf{Distance fallen: } \Delta y \approx 5t^2 \textbf{ (meters)} \qquad \textbf{Speed acquired: } v \approx 10t \textbf{ (m/s)}$$

Memorize the Integer Time Progression: • $t = 1$ s → $\Delta y = 5$ m, $v = 10$ m/s • $t = 2$ s → $\Delta y = 20$ m, $v = 20$ m/s • $t = 3$ s → $\Delta y = 45$ m, $v = 30$ m/s • $t = 4$ s → $\Delta y = 80$ m, $v = 40$ m/s • $t = 5$ s → $\Delta y = 125$ m, $v = 50$ m/s (180 km/h!)

Precision Adjustment: Since 9.80 is exactly 2% smaller than 10, subtract 2% from your mental estimate: For $t = 3$ s: 45 m − 2%(45) = 45 − 0.9 = **44.1 m** (Exact: ½ × 9.8 × 9 = 44.1 m!). Solved in 2 seconds!

TRICK K2 Galileo's Odd-Number Law for Successive Equal Time Intervals

Ratio Rule

Principle: For any body accelerating uniformly from rest ($v_0 = 0$), the distances traversed during successive equal time intervals Δt follow the ratio of successive odd numbers:

$$\Delta x_1 : \Delta x_2 : \Delta x_3 : \Delta x_4 : \ldots = 1 : 3 : 5 : 7 : \ldots : (2n - 1)$$

Instant Application: If a car accelerates from rest and travels 4 meters in the 1st second, it travels 4 × 3 = **12 m** in the 2nd second, 4 × 5 = **20 m** in the 3rd second, and 4 × 7 = **28 m** in the 4th second. Total displacement after 4 seconds = $4 \times (4^2)$ = **64 m**!

Part II: Two-Dimensional Kinematics & Projectile Motion

Chapter 4: Orthogonal Motion Independence & 2D Vector Equations

The central pillar of multidimensional Newtonian mechanics is **Galileo's Principle of Independence of Perpendicular Motions**: an acceleration or force acting strictly along one spatial axis exerts zero influence on the kinematic state along any perpendicular axis. When a particle moves in the *xy*-plane under constant acceleration $\mathbf{a} = a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}}$, the full 2D motion decomposes into two completely independent, simultaneous 1D constant-acceleration problems:

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{a}\,t^2 \quad \Leftrightarrow \quad \begin{aligned} x(t) &= x_0 + v_{0x}t + \tfrac{1}{2}a_x t^2 \\ y(t) &= y_0 + v_{0y}t + \tfrac{1}{2}a_y t^2 \end{aligned}$$

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}t \quad \Leftrightarrow \quad \begin{aligned} v_x(t) &= v_{0x} + a_x t \\ v_y(t) &= v_{0y} + a_y t \end{aligned}$$

Chapter 5: Projectile Motion Analytical Theory: Time, Apex, and Range

Projectile motion describes the unpowered trajectory of an object launched into a uniform gravitational field in the absence of atmospheric drag. Choosing the coordinate origin at the launch site, with positive *x* directed horizontally forward and positive *y* directed vertically upward, the gravitational acceleration vector is strictly:

$$\mathbf{a} = 0\hat{\mathbf{i}} - g\hat{\mathbf{j}} \quad \Rightarrow \quad a_x = 0, \quad a_y = -g = -9.80\ \text{m/s}^2$$

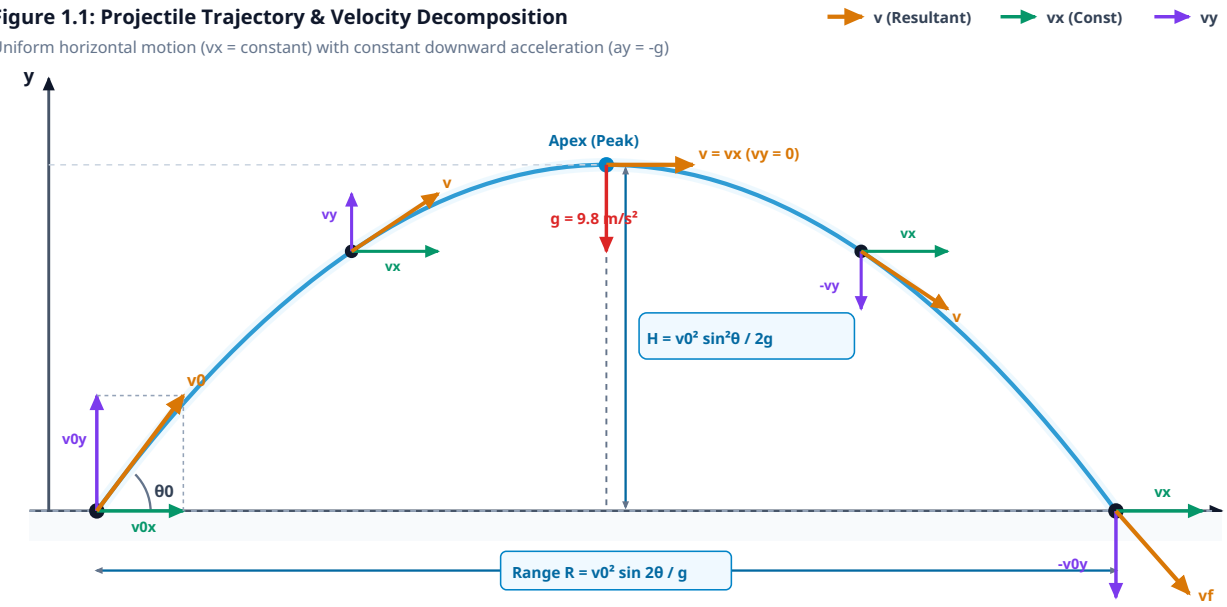
If the projectile is launched with initial speed v_0 at an inclination angle θ above the horizontal, the initial velocity resolves into orthogonal components:

$$v_{0x} = v_0 \cos\theta, \qquad v_{0y} = v_0 \sin\theta$$

Because $a_x = 0$, the horizontal velocity remains strictly invariant throughout the entire flight: $v_x(t) = v_{0x} = v_0 \cos\theta = \text{const}$. Meanwhile, the vertical motion is governed by 1D free-fall: $v_y(t) = v_{0y} - gt$.

Figure 1.1: Projectile Trajectory & Velocity Decomposition

Uniform horizontal motion (vx = constant) with constant downward acceleration (ay = -g)



5.1 Derivation of Key Trajectory Flight Parameters (Flat Ground)

1. Time to Apex (Peak) t_{apex}: At the highest point of the trajectory, the vertical velocity momentarily vanishes: $v_y(t_{apex}) = 0$.

$$0 = v_{0y} - g\,t_{apex} \quad \Rightarrow \quad \boldsymbol{t_{apex} = \frac{v_{0y}}{g} = \frac{v_0 \sin\theta}{g}}$$

2. Maximum Apex Height *H*: Using the time-independent kinematic relation $v_y^2 = v_{0y}^2 - 2gH$ with $v_y = 0$:

$$0 = v_0^2 \sin^2\theta - 2gH \quad \Rightarrow \quad \boldsymbol{H = \frac{v_{0y}^2}{2g} = \frac{v_0^2 \sin^2\theta}{2g}}$$

3. Total Time of Flight *T*: For level ground ($y(T) = y_0 = 0$), vertical displacement is zero: $0 = v_{0y}T - \tfrac{1}{2}gT^2 = T(v_{0y} - \tfrac{1}{2}gT)$. Discarding the trivial launch root $T = 0$ yields:

$$\boldsymbol{T = 2\,t_{apex} = \frac{2\,v_{0y}}{g} = \frac{2\,v_0 \sin\theta}{g}}$$

4. Horizontal Range *R*: The horizontal displacement traversed over the full flight duration T is:

$$R = v_{0x}T = (v_0\cos\theta)\,\frac{2\,v_0\sin\theta}{g} = \frac{v_0^2\,(2\sin\theta\cos\theta)}{g}$$

Applying the double-angle trigonometric identity $2\sin\theta\cos\theta = \sin 2\theta$, we establish the standard Range Formula:

$$R = \frac{v_0^2 \sin 2\theta}{g}$$

TRICK K3 **The Universal Projectile Range-to-Height Identity: R = 4H cot θ** **Instant Ratio**

Derivation: Take the ratio of the horizontal range R to the maximum height H:

$$\frac{R}{H} = \frac{(v_0^2 / g)(2 \sin\theta \cos\theta)}{(v_0^2 / 2g)(\sin^2\theta)} = \frac{4\cos\theta}{\sin\theta} = \mathbf{4\cot\theta} \quad \Leftrightarrow \quad \mathbf{\tan\theta = \frac{4H}{R}}$$

Three Powerful Takeaways for Exams:

1. **For θ = 45°:** cot 45° = 1 ⇒ $\boldsymbol{R_{max} = 4H}$. The maximum range is always exactly four times the apex height!
2. **When Range Equals Height ($R = H$):** tan θ = 4 ⇒ θ = arctan(4) ≈ **75.96°**.
3. **Zero-Equation Launch Angle Identification:** If an artillery shell lands 800 m away with an apex of 200 m, tan θ = 4(200)/800 = 1.0 ⇒ θ = **45.0°** instantly!

Chapter 6: Trajectory Parabola, Launch Angle Symmetry, & Asymmetric Elevations

6.1 The Cartesian Trajectory Equation $y(x)$

To determine the spatial geometry of the flight path independent of the parameter t, we solve the horizontal position equation for time: $t = x / (v_0 \cos\theta)$. Substituting this expression into the vertical position equation $y(t)$:

$$y(x) = (v_0 \sin\theta)\frac{x}{v_0\cos\theta} - \frac{1}{2} g \frac{x^2}{v_0^2\cos^2\theta} = \mathbf{(\tan\theta)\,x - \frac{g}{2v_0^2\cos^2\theta}\,x^2}$$

This is an inverted quadratic equation of the canonical form $y(x) = Ax - Bx^2$, proving rigorously that **all projectile paths under uniform gravity are true mathematical parabolas**. Using the trigonometric identity $1 / \cos^2\theta = 1 + \tan^2\theta = \sec^2\theta$, the trajectory equation can be recast in terms of tan θ:

$$y(x) = (\tan\theta)\,x - \frac{g\,x^2}{2v_0^2}(1 + \tan^2\theta)$$

6.2 Complementary Launch Angles and Maximum Range

For a given initial speed v_0 on flat ground, the horizontal range $R = (v_0^2 / g) \sin 2\theta$ is maximized when sin 2θ reaches its maximum value of 1:

$$\sin 2\theta = 1 \quad \Rightarrow \quad 2\theta = 90° \quad \Rightarrow \quad \mathbf{\theta_{optimal} = 45°}, \quad \mathbf{R_{max} = \frac{v_0^2}{g}}$$

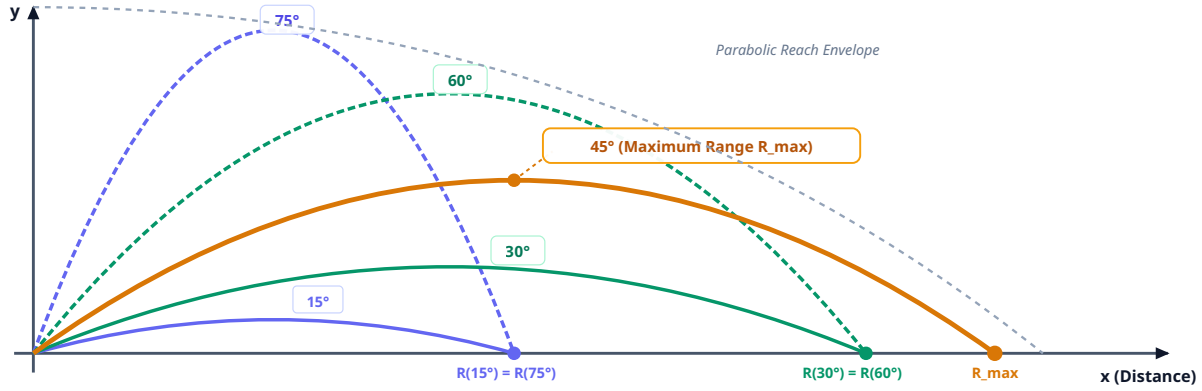
Furthermore, consider two complementary launch angles $\theta_1 = \theta$ and $\theta_2 = 90° - \theta$. Evaluating $\sin 2\theta_2$:

$$\sin(2(90° - \theta)) = \sin(180° - 2\theta) = \sin 2\theta$$

Therefore, **any pair of complementary launch angles produces the identical horizontal range *R***! While their ranges match, their flight times and maximum heights differ drastically: the steeper angle spends far more time in the air and reaches a much higher peak.

Figure 1.4: Launch Angle Trajectory Comparison & Complementary Range Symmetry

Identical horizontal range R(θ) = R(90° − θ) for complementary pairs with identical launch speed v0



TRICK K4 **Trajectory Speed at Any Elevation without Time (Energy-Kinematics Shortcut)** **5 Seconds**

Standard Method (Slow): Solve the quadratic equation for time t when $y(t) = h$, then calculate $v_y = v_{0y} - gt$, and finally compute $v = \sqrt{(v_x^2 + v_y^2)}$. (Requires 2–3 minutes).

⚡ **Fast Hand Calculation Shortcut:** Use the work-energy kinematic equivalence:

$$v(y) = \sqrt{(v_0^2 - 2g\,\Delta y)}$$

Vital Insight: The speed v at any vertical height y depends **strictly on the vertical displacement Δy** and is completely independent of the launch angle θ or horizontal distance!

Example: Launch at $v_0 = 20$ m/s at *any* angle. What is the speed when it is 15 m above launch? $v = \sqrt{(20^2 - 2(9.8)(15))} = \sqrt{(400 - 294)} = \sqrt{106} \approx$ **10.3 m/s**. Solved in 5 seconds!

Part III: Uniform and Non-Uniform Circular Kinematics

Chapter 7: Circular Coordinate Variables, Angular Speed, and Frequency

When a particle is constrained to travel along a circular path of fixed radius R, Cartesian coordinates (x, y) become redundant because the radial distance is constant: $x^2 + y^2 = R^2$. Motion is most naturally parameterized by the scalar **angular displacement** $\theta(t)$ measured in radians:

$$x(t) = R\cos\theta(t), \qquad y(t) = R\sin\theta(t)$$

The relationship between linear arc length s traversed along the perimeter and angular displacement θ is:

$$\boldsymbol{s = R\,\theta} \quad (\theta \text{ in radians})$$

Differentiating with respect to time yields the continuous angular and linear kinematic variables:

- **Angular Velocity ω:** The time rate of change of angular position:

$$\omega = \frac{d\theta}{dt} \quad \text{(rad/s)} \quad \Leftrightarrow \quad \boldsymbol{v = \omega R}$$

- **Angular Acceleration α:** The time rate of change of angular velocity:

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \quad \text{(rad/s}^2\text{)} \quad \Leftrightarrow \quad \boldsymbol{a_t = \alpha R}$$

- **Period T and Frequency f:** For uniform circular motion with constant ω:

$$T = \frac{2\pi}{\omega} = \frac{2\pi R}{v}, \qquad f = \frac{1}{T} = \frac{\omega}{2\pi} \quad \text{(Hz or rev/s)}$$

Chapter 8: Centripetal (Radial) Acceleration Geometric & Vector Proofs

Even when a particle traverses a circular path at constant speed ($|\mathbf{v}|$ = const, uniform circular motion), its velocity vector $\mathbf{v}$ is continuously changing direction. Since acceleration is the derivative of the velocity vector, **uniform circular motion is fundamentally accelerated motion**.

8.1 Rigorous Unit Vector Derivation of Centripetal Acceleration

In polar coordinates, we define the radial unit vector $\hat{\mathbf{r}}(t) = \cos\theta\,\hat{\mathbf{i}} + \sin\theta\,\hat{\mathbf{j}}$ and the transverse tangential unit vector $\hat{\boldsymbol{\theta}}(t) = -\sin\theta\,\hat{\mathbf{i}} + \cos\theta\,\hat{\mathbf{j}}$. Differentiating these rotating unit vectors with respect to time using the chain rule:

$$\frac{d\hat{\mathbf{r}}}{dt} = (-\sin\theta\,\hat{\mathbf{i}} + \cos\theta\,\hat{\mathbf{j}})\,\frac{d\theta}{dt} = \omega\,\hat{\boldsymbol{\theta}}, \qquad \frac{d\hat{\boldsymbol{\theta}}}{dt} = (-\cos\theta\,\hat{\mathbf{i}} - \sin\theta\,\hat{\mathbf{j}})\,\frac{d\theta}{dt} = -\omega\,\hat{\mathbf{r}}$$

The position vector is $\mathbf{r} = R\,\hat{\mathbf{r}}$. The velocity vector is:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = R\,\frac{d\hat{\mathbf{r}}}{dt} = R\,\omega\,\hat{\boldsymbol{\theta}} = v\,\hat{\boldsymbol{\theta}}$$

Differentiating once more to find acceleration:

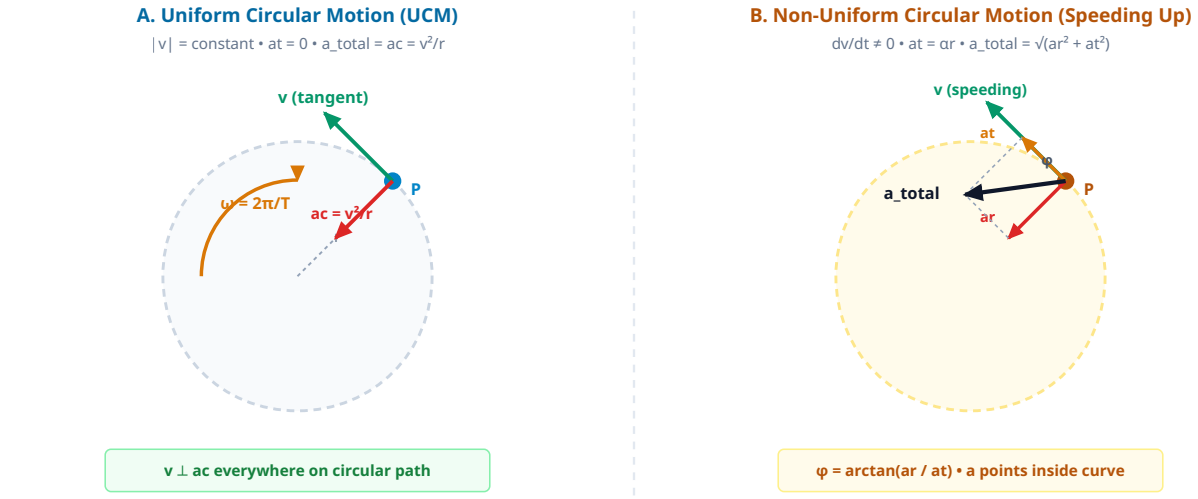
$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{dv}{dt}\,\hat{\boldsymbol{\theta}} + v\,\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{dv}{dt}\,\hat{\boldsymbol{\theta}} + v\,(-\omega\,\hat{\mathbf{r}}) = -\omega^2\,R\,\hat{\mathbf{r}} + \frac{dv}{dt}\,\hat{\boldsymbol{\theta}}$$

The inward radial component is the celebrated **centripetal acceleration** a_c:

$$a_c = \frac{v^2}{R} = \omega^2 R = \frac{4\pi^2 R}{T^2}$$

Figure 1.2: Circular Kinematics — Uniform vs. Non-Uniform Acceleration

Comparison of acceleration vectors in uniform (pure radial ac) versus non-uniform motion (radial ar + tangential at)



Chapter 9: Non-Uniform Circular Kinematics, Tangential Acceleration, & Curvature

If the particle's speed changes along the circular track ($dv/dt \neq 0$), the motion is **non-uniform circular motion**. The net acceleration vector **a** consists of two mutually perpendicular components:

1. **Radial (Centripetal) Acceleration a_r:** Governs the change in velocity *direction*: $a_r = v^2 / R$. Directed toward the instantaneous center of curvature.

2. **Tangential Acceleration a_t:** Governs the change in velocity *magnitude* (speed): $a_t = dv/dt = \alpha R$. Directed tangent to the circular trajectory.

Because $\mathbf{a}_r \perp \mathbf{a}_t$, the total acceleration magnitude is given by the Pythagorean theorem:

$$a_{\text{total}} = \sqrt{(a_r^2 + a_t^2)} = \sqrt{\left[\left(\frac{v^2}{R} \right)^2 + (\alpha R)^2 \right]}$$

The angle φ of the total acceleration vector relative to the forward tangential direction is:

$$\varphi = \arctan \frac{a_r}{a_t} = \arctan \frac{v^2}{R\, a_t}$$

TRICK K5 **Circular Acceleration Scaling Laws & The Equal-Acceleration Angle** **Ratio Magic**

Three Mental Ratio Rules for Centripetal Acceleration:

• **Constant Radius (R fixed):** $a_c \propto v^2 \Rightarrow$ If speed doubles, centripetal acceleration **quadruples (4×)**.

• **Constant Linear Speed (v fixed):** $a_c \propto 1/R \Rightarrow$ If curve radius doubles, acceleration **halves (½×)**.

• **Constant Angular Speed (ω fixed, e.g., spinning disk):** $a_c \propto R \Rightarrow$ An insect twice as far from the center experiences **double (2×)** the acceleration.

⚡ **The Golden Equal-Acceleration Angle (θ* = ½ rad):**

For any wheel or vehicle accelerating from rest ($\omega_0 = 0$) with constant angular acceleration α:

$\omega^2 = 2\alpha\theta \quad \Rightarrow \quad a_r = \omega^2 R = 2\alpha\theta R = 2\theta\, a_t$.

Centripetal acceleration equals tangential acceleration ($a_r = a_t$) when $2\theta = 1$:

$$\theta^* = \frac{1}{2} \text{ rad} \approx 28.65° \approx 0.0796 \text{ revolutions}$$

This universal angle holds **regardless of radius R or acceleration α**!

Part IV: Relative Motion & Reference Frames

Chapter 10: Inertial Reference Frames & Galilean Position Transformations

A **frame of reference** is a physical coordinate system anchored to an observer equipped with a synchronized clock. An **inertial reference frame** is defined as one in which Newton's first law of motion holds true: a free particle experiences zero acceleration in the absence of net external forces. Any frame moving with constant velocity relative to an inertial frame is likewise an inertial frame.

Let Frame A and Frame B be two reference frames, where Frame B translates with velocity $\mathbf{v}_{B/A}$ relative to Frame A. The position vector of a particle P measured by observer A is related to the position vector measured by observer B by the **Galilean Position Transformation**:

$$\mathbf{r}_{P/A} = \mathbf{r}_{P/B} + \mathbf{r}_{B/A}$$

Chapter 11: Galilean Velocity Addition: River Crossing & Aerial Crosswind Vectors

Differentiating the position transformation with respect to time t (assuming universal time $t = t'$ in classical physics):

$$\frac{d\mathbf{r}_{P/A}}{dt} = \frac{d\mathbf{r}_{P/B}}{dt} + \frac{d\mathbf{r}_{B/A}}{dt}$$

This establishes the classical **Galilean Velocity Addition Theorem**:

$$\mathbf{v}_{P/A} = \mathbf{v}_{P/B} + \mathbf{v}_{B/A}$$

Mnemonic Subscript Rule: The intermediate reference frame subscript cancels in cyclic matrix fashion: $\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C}$. Reversing relative subscripts introduces a sign negation: $\mathbf{v}_{B/A} = -\mathbf{v}_{A/B}$.

11.1 Frame Invariance of Acceleration

Differentiating the velocity relation with respect to time:

$$\mathbf{a}_{P/A} = \mathbf{a}_{P/B} + \mathbf{a}_{B/A}$$

If Frame B is non-accelerating (translating at constant velocity relative to Frame A, so $\mathbf{a}_{B/A} = \mathbf{0}$):

$$\mathbf{a}_{P/A} = \mathbf{a}_{P/B}$$

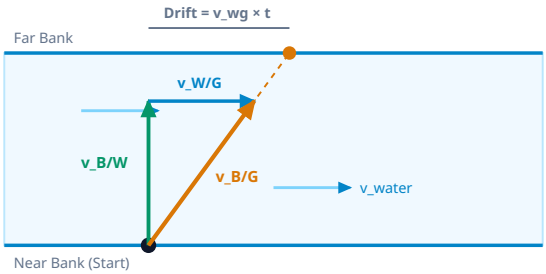
Fundamental Theorem: *All inertial observers measure identical accelerations for any moving particle.* Consequently, Newton's second law $\mathbf{F} = m\mathbf{a}$ is invariant across all inertial reference frames!

Figure 1.3: 2D Relative Velocity & River Crossing Vector Mechanics

Vector addition v_boat/ground = v_boat/water + v_water/ground under Galilean transformation

Case A: Minimum Crossing Time

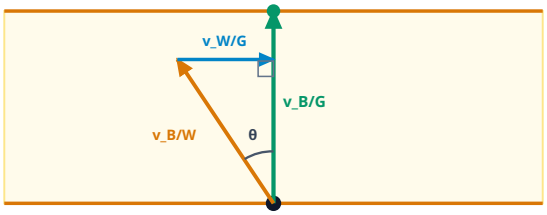
Heading Perpendicular to Banks • t_min = d / v_bw



t = d / v_bw (Independent of current speed!)

Case B: Zero Drift (Shortest Path)

Heading Upstream at Angle θ • Net Velocity Due North



sin θ = v_wg / v_bw • v_bg = √(v_bw² − v_wg²)

TRICK K6 **River Crossing Navigation Shortcut: Minimum Time vs. Zero Drift** **10 Seconds**

A boat with motor speed $v_{B/W}$ crosses a river of width d flowing with current speed $v_{W/G}$.

Scenario 1: Shortest Crossing Time (Minimum Time):

- **Heading:** Aim straight across (⊥ to banks). The full boat engine power drives across the river.
- **Time:** $t_{min} = \frac{d}{v_{B/W}}$ (Completely independent of how fast the river flows!).
- **Downstream Drift:** $x_{drift} = v_{W/G} \times t_{min}$.

Scenario 2: Shortest Distance (Zero Drift / Straight Across):

- **Condition:** Requires $v_{B/W} \geq v_{W/G}$ (boat must be faster than current).
- **Upstream Angle:** $\sin\theta = \frac{v_{W/G}}{v_{B/W}}$.
- **Resultant Groundspeed:** $v_{B/G} = \sqrt{(v_{B/W}^2 - v_{W/G}^2)}$.
- **Crossing Time:** $t = \frac{d}{\sqrt{(v_{B/W}^2 - v_{W/G}^2)}}$. Always strictly longer than t_{min}!

Part V: Step-by-Step Worked Problems with Fast-Tracks

Worked Problem 1: Level-Ground Projectile Motion with Vector Decomposition

Lecture Slides Ex 1.2 • Fast-Track K3

1. PROBLEM STATEMENT

A football placekicker kicks a football from ground level at an angle of θ = 40.0° above the horizontal with an initial velocity of v_0 = 22.0 m/s. Neglecting air resistance and taking g = 9.80 m/s^2:

(a) Determine the maximum height H attained by the ball.
(b) Calculate the total time of flight T between kickoff and landing.
(c) Calculate the horizontal range R of the projectile.
(d) Determine the landing velocity vector $\mathbf{v}_f$.

2. GIVEN PARAMETERS & KNOWNS

• Initial speed: v_0 = 22.0 m/s • Launch angle: θ = 40.0° • Initial position: x_0 = 0 m, y_0 = 0 m

• Gravitational acceleration: a_x = 0 m/s^2, $a_y = -g$ = −9.80 m/s^2

3. SYMBOLIC ALGEBRAIC DERIVATIONS

Initial velocity components:

$v_{0x} = v_0 \cos\theta$, $v_{0y} = v_0 \sin\theta$

(a) At apex, $v_y = 0 \Rightarrow v_y^2 = v_{0y}^2 - 2gH = 0 \Rightarrow \boldsymbol{H = v_0^2 \sin^2\theta / (2g)}$

(b) Flight time to landing (y = 0): $0 = v_{0y}T - \tfrac{1}{2}gT^2 \Rightarrow \boldsymbol{T = 2v_0 \sin\theta / g}$

(c) Range: $\boldsymbol{R = v_{0x}T = v_0^2 \sin 2\theta / g}$

(d) Landing velocity: $v_{fx} = v_{0x}$, $v_{fy} = -v_{0y} \Rightarrow \mathbf{v}_f = v_{0x}\hat{\mathbf{i}} - v_{0y}\hat{\mathbf{j}}$

4. NUMERICAL EVALUATION WITH SI UNITS

• $v_{0x} = 22.0 \times \cos(40.0°) = 22.0 \times 0.76604 =$ **16.85 m/s**

• $v_{0y} = 22.0 \times \sin(40.0°) = 22.0 \times 0.64279 =$ **14.14 m/s**

• **(a) Maximum Height:** $H = (14.141)^2 / (2 \times 9.80) = 199.97 / 19.60 =$ **10.20 m**
• **(b) Total Flight Time:** $T = 2(14.141) / 9.80 = 28.282 / 9.80 =$ **2.886 s ≈ 2.89 s**
• **(c) Horizontal Range:** $R = 16.853 \times 2.886 =$ **48.64 m ≈ 48.6 m**
  (Double check via formula: $(22.0)^2 \sin(80.0°) / 9.80 = (484 \times 0.9848) / 9.80 = 48.64$ m)
• **(d) Impact Velocity Vector:** $\mathbf{v}_f =$ **16.85 î − 14.14 ĵ m/s** ($|\mathbf{v}_f|$ = **22.0 m/s** at −40.0°)

Physical Sanity Check via Fast-Track K3:
Evaluate the ratio R / H = 48.64 / 10.20 = **4.768**.
Check against theoretical prediction: 4 cot(40.0°) = 4 / tan(40.0°) = 4 / 0.83910 = **4.767**.
Exact numerical agreement confirms complete mathematical consistency!

Worked Problem 2: Asymmetric Cliff Launch with Deep Canyon Impact

Textbook Shankar Ch 2 • Fast-Track K4

1. PROBLEM STATEMENT

An emergency rescue team launches a supply canister from the edge of a vertical cliff of height h = 100.0 m with an initial velocity of v_0 = 20.0 m/s angled at θ = 30.0° above the horizontal. Air drag is negligible and g = 9.80 m/s^2. Determine:

(a) The total flight duration t_{land} until the canister impacts the canyon floor.

(b) The horizontal distance x_{land} from the base of the cliff to the impact point.

(c) The velocity vector components and resultant impact speed v_{impact}.

(d) The angle of impact β relative to the horizontal canyon floor.

2. GIVEN PARAMETERS & KNOWNS

• Cliff elevation: y_0 = 0 m (set cliff top as origin), landing elevation y_{land} = −100.0 m

• Initial speed: v_0 = 20.0 m/s • Angle: θ = 30.0° • g = 9.80 m/s^2

3. SYMBOLIC ALGEBRAIC DERIVATIONS

$v_{0x} = v_0 \cos\theta$, $v_{0y} = v_0 \sin\theta$

(a) Vertical trajectory: $y(t) = v_{0y}t - \tfrac{1}{2}gt^2 = -h$ ⇒ $\mathbf{\tfrac{1}{2}gt^2 - v_{0y}t - h = 0}$

Applying the quadratic formula for positive time root: $\mathbf{t_{land} =}$

$$\frac{v_{0y} + \sqrt{(v_{0y}^2 + 2gh)}}{g}$$

(b) Horizontal distance: $\mathbf{x_{land} = v_{0x}\, t_{land}}$

(c) Vertical velocity at landing: $\mathbf{v_{fy} = v_{0y} - g\, t_{land} = -\sqrt{(v_{0y}^2 + 2gh)}}$

Total speed: $\mathbf{v_{impact} = \sqrt{(v_{fx}^2 + v_{fy}^2)} = \sqrt{(v_0^2 + 2gh)}}$

4. NUMERICAL EVALUATION WITH SI UNITS

• v_{0x} = 20.0 cos(30.0°) = 20.0 × 0.86603 = **17.32 m/s**

• v_{0y} = 20.0 sin(30.0°) = 20.0 × 0.50000 = **10.00 m/s**

• Discriminant term: √[10.0^2 + 2(9.80)(100.0)] = √[100.0 + 1960.0] = √2060.0 = **45.387 m/s**

• **(a) Flight Time:** t_{land} = (10.00 + 45.387) / 9.80 = 55.387 / 9.80 = **5.652 s ≈ 5.65 s**

• **(b) Horizontal Distance:** x_{land} = 17.3205 × 5.6518 = **97.89 m ≈ 97.9 m**

• **(c) Impact Velocity:** v_{fx} = 17.32 m/s, v_{fy} = −45.39 m/s.

v_{impact} = √[17.3205^2 + (−45.387)^2] = √[300.0 + 2060.0] = √2360.0 = **48.58 m/s**

• **(d) Impact Angle:** β = arctan(|v_{fy}| / v_{fx}) = arctan(45.387 / 17.3205) = arctan(2.6204) = **69.1° below horizontal**

Physical Sanity Check via Fast-Track K4 (Energy Conservation):

v_{impact} = √(v_0^2 + 2gh) = √(20.0^2 + 2(9.80)(100.0)) = √(400.0 + 1960.0) = √2360.0 = **48.58 m/s**.

Matches kinematic evaluation to four decimal digits. Notice also that since t_{land} ≈ 5.65 s > 2t_{apex} = 2(10/9.8) ≈ 2.04 s, the canister is confirmed to land on the downward falling phase.

Worked Problem 3: Tennis Ball Hit into Elevated Stadium Grandstand

Lecture Slides Arthur Ashe Stadium Example

1. PROBLEM STATEMENT

A champion tennis player strikes a match-winning ball at v_0 = 30.0 m/s angled at θ = 45.0° above the horizontal. On its way downward, the ball is caught by a spectator in the grandstand at an elevation y_{catch} = 10.0 m above the launch height.

(a) Calculate the time elapsed t_{catch} when the spectator catches the ball.

(b) Find the horizontal distance x_{catch} from the baseline to the spectator.

(c) Compute the velocity vector **v**, magnitude, and impact angle at the instant of capture.

2. GIVEN PARAMETERS & KNOWNS

• v_0 = 30.0 m/s • θ = 45.0° • y_{catch} = +10.0 m • g = 9.80 m/s^2

• v_{0x} = 30.0 cos(45°) = 21.213 m/s, v_{0y} = 30.0 sin(45°) = 21.213 m/s

3. SYMBOLIC ALGEBRAIC DERIVATIONS

(a) Vertical trajectory quadratic: $y(t) = v_{0y}t - \frac{1}{2}gt^2 = h$ ⇒ $\mathbf{4.90\,t^2 - v_{0y}\,t + h = 0}$

Discriminant: $D = v_{0y}^2 - 4(4.90)h = v_{0y}^2 - 2gh$.

Two physical roots exist: t_1 (ascending pass) and t_2 (descending pass). Because the problem explicitly states the ball is caught **on its way down**, we select the larger root: $\boldsymbol{t_{catch} = (v_{0y} + \sqrt{(v_{0y}^2 - 2gh)})\,/\,g}$

(b) $\boldsymbol{x_{catch} = v_{0x}\,t_{catch}}$

(c) $v_x = v_{0x}$, $\boldsymbol{v_y = v_{0y} - g\,t_{catch} = -\sqrt{(v_{0y}^2 - 2gh)}}$

4. NUMERICAL EVALUATION WITH SI UNITS

• $v_{0y}^2 = (21.2132)^2 = 450.0$ m^2/s^2

• $2gh = 2(9.80)(10.0) = 196.0$ m^2/s^2

• √[450.0 − 196.0] = √254.0 = **15.937 m/s**

• Ascent root: t_1 = (21.213 − 15.937) / 9.80 = 5.276 / 9.80 = 0.538 s (ball passing upward through 10 m)

• **(a) Descent Catch Time:** t_{catch} = (21.213 + 15.937) / 9.80 = 37.150 / 9.80 = **3.791 s ≈ 3.79 s**

• **(b) Horizontal Distance:** x_{catch} = 21.2132 × 3.7908 = **80.42 m ≈ 80.4 m**

• **(c) Impact Velocity:** v_x = **21.21 m/s**, v_y = **−15.94 m/s**

Magnitude: v = √[21.213^2 + (−15.937)^2] = √[450.0 + 254.0] = √704.0 = **26.53 m/s**

Angle: θ_{catch} = arctan(−15.937 / 21.213) = **−36.9° (36.9° below horizontal)**

Physical Sanity Check:

Apex height is $H = v_{0y}^2 / 2g$ = 450 / 19.6 = **22.96 m**. Since 10.0 m < 22.96 m, the ball easily cleared the required height.

Via Fast-Track K4: $v = \sqrt{(v_0^2 - 2gh)}$ = √(900 − 196) = √704 = **26.53 m/s**! Exact match!

Worked Problem 4: Non-Uniform Circular Kinematics of a Supercar

Textbook & Lecture Slides Circular Motion • Fast-Track K5

1. PROBLEM STATEMENT

A high-performance prototype car enters a circular test track of radius R = 120.0 m with an initial speed of v_0 = 15.0 m/s. The driver depresses the throttle to produce a constant tangential acceleration of a_t = 2.50 m/s^2.

(a) Compute the angular acceleration α of the vehicle.
(b) Find the time t^* and distance s^* at which centripetal acceleration equals tangential acceleration.
(c) Find the elapsed time t_{total} when the total acceleration magnitude reaches a_{net} = 5.00 m/s^2.
(d) Determine the total arc length s and number of revolutions completed during this time.

2. GIVEN PARAMETERS & KNOWNS

• Track radius: R = 120.0 m • Initial speed: v_0 = 15.0 m/s
• Constant tangential acceleration: a_t = 2.50 m/s^2 • Target net acceleration: a_{net} = 5.00 m/s^2

3. SYMBOLIC ALGEBRAIC DERIVATIONS

(a) Angular acceleration: $\boldsymbol{\alpha = a_t / R}$

(b) Condition $a_r = a_t \Rightarrow v^2 / R = a_t \Rightarrow v^* = \sqrt{(a_t R)}$. Using $v(t) = v_0 + a_t t \Rightarrow \boldsymbol{t^* = (\sqrt{(a_t R)} - v_0) / a_t}$

(c) For net acceleration $a_{net} = \sqrt{(a_r^2 + a_t^2)} \Rightarrow \boldsymbol{a_r = \sqrt{(a_{net}^2 - a_t^2)}}$

From $a_r = v^2 / R \Rightarrow v_{final} = \sqrt{(R\, a_r)} \Rightarrow \boldsymbol{t_{total} = (v_{final} - v_0) / a_t}$

(d) Kinematic arc length: $\boldsymbol{s = v_0 t_{total} + \tfrac{1}{2} a_t t_{total}^2}$, Revolutions: $\boldsymbol{N = s / (2\pi R)}$

4. NUMERICAL EVALUATION WITH SI UNITS

• **(a) Angular Acceleration:** α = 2.50 / 120.0 = **0.02083 rad/s^2 ≈ 2.08 × 10^{-2} rad/s^2**
• **(b) Time for $a_r = a_t$:** v^* = √(2.50 × 120.0) = √300.0 = 17.321 m/s.
  t^* = (17.321 − 15.0) / 2.50 = 2.321 / 2.50 = **0.928 s**
  Arc length: s^* = ($v^{*2} - v_0^2$) / (2a_t) = (300.0 − 225.0) / 5.00 = **15.00 m**
• **(c) Time for a_{net} = 5.00 m/s^2:**
  a_r = √[5.00^2 − 2.50^2] = √[25.0 − 6.25] = √18.75 = **4.330 m/s^2**
  v_{final} = √[120.0 × 4.3301] = √519.615 = **22.795 m/s** (82.1 km/h)
  t_{total} = (22.795 − 15.0) / 2.50 = 7.795 / 2.50 = **3.118 s ≈ 3.12 s**
• **(d) Arc Length & Revolutions:**
  s = (22.795^2 − 15.0^2) / (2 × 2.50) = (519.62 − 225.0) / 5.00 = **58.92 m**
  Track circumference = 2π(120.0) = 753.98 m ⇒ N = 58.92 / 753.98 = **0.0781 rev (28.1°)**

Physical Sanity Check:
At t = 3.118 s, v = 15.0 + 2.50(3.118) = 22.795 m/s.
Check total acceleration: a_r = 22.795^2 / 120.0 = 519.615 / 120.0 = 4.330 m/s^2.
a_{net} = √(2.50^2 + 4.330^2) = √(6.25 + 18.75) = √25.00 = **5.000 m/s^2**. Exact match!

Worked Problem 5: 2D Relative Velocity in Crosswind Aerial Navigation

Galilean Velocity • Fast-Track K6

1. PROBLEM STATEMENT

A twin-engine transport aircraft is scheduled to fly due North from City A to City B, separated by a direct distance of D = 500.0 km. The aircraft's cruising airspeed (speed relative to air mass) is $v_{P/A}$ = 250.0 km/h. A persistent regional weather front produces a steady crosswind blowing from the West toward the East at $v_{A/G}$ = 65.0 km/h.

(a) Calculate the heading angle θ (degrees West of North) the pilot must steer so that the ground track is due North.

(b) Determine the resultant groundspeed $v_{P/G}$ of the aircraft.

(c) Find the total flight time required to complete the journey.

(d) *Navigation Error Assessment:* If the pilot pointed the nose due North without compensating for the crosswind, calculate the groundspeed and the lateral drift error Δx when the plane reaches the latitude of City B.

2. GIVEN PARAMETERS & KNOWNS

• Distance: D = 500.0 km due North • Airspeed: $v_{P/A}$ = 250.0 km/h

• Wind velocity: $\mathbf{v}_{A/G}$ = +65.0 $\hat{\mathbf{i}}$ km/h (blowing Eastward) • Desired track: $\mathbf{v}_{P/G} = 0\,\hat{\mathbf{i}} + v_{P/G}\,\hat{\mathbf{j}}$

3. SYMBOLIC ALGEBRAIC DERIVATIONS

Galilean velocity vector relation: $\mathbf{v}_{P/G} = \mathbf{v}_{P/A} + \mathbf{v}_{A/G}$

Resolving into components with heading θ West of North:

$\mathbf{v}_{P/A} = (-v_{P/A} \sin\theta)\,\hat{\mathbf{i}} + (v_{P/A} \cos\theta)\,\hat{\mathbf{j}}$

(a) To maintain a due North track, the net East-West velocity must vanish:

$-v_{P/A} \sin\theta + v_{A/G} = 0 \quad \Rightarrow \quad \mathbf{\sin\theta = \frac{v_{A/G}}{v_{P/A}}} \quad \Rightarrow \quad \boldsymbol{\theta = \arcsin(v_{A/G} / v_{P/A})}$

(b) Resultant groundspeed along North: $\boldsymbol{v_{P/G} = v_{P/A} \cos\theta = \sqrt{(v_{P/A}^2 - v_{A/G}^2)}}$

(c) Flight duration: $\boldsymbol{t = D / v_{P/G}}$

(d) Uncompensated flight: $\mathbf{v}' = 65.0\,\hat{\mathbf{i}} + 250.0\,\hat{\mathbf{j}}$ km/h. Time to latitude B: $t' = D / 250.0$. Drift: $\boldsymbol{\Delta x = v_{A/G}\, t'}$

4. NUMERICAL EVALUATION WITH SI UNITS

• **(a) Heading Angle:** $\sin\theta = 65.0 / 250.0 = 0.2600 \Rightarrow \theta = \arcsin(0.2600) =$ **15.07° ≈ 15.1° West of North (Heading 344.9°)**

• **(b) Resultant Groundspeed:** $v_{P/G} = \sqrt{[250.0^2 - 65.0^2]} = \sqrt{[62500 - 4225]} = \sqrt{58275} =$ **241.40 km/h = 67.06 m/s**

• **(c) Flight Duration:** t = 500.0 km / 241.40 km/h = **2.0712 hours = 2 h 4 min 16 s**

• **(d) Navigation Error if Uncompensated:** v' = √[250.0² + 65.0²] = 258.31 km/h. Time to latitude B: t' = 2.000 h. Lateral Drift: Δx = 65.0 × 2.000 = **130.0 km East of City B!**

Physical Sanity Check:

Verify Pythagoras identity: $\sin^2\theta + \cos^2\theta = (65/250)^2 + (241.40/250)^2 = 0.06760 + 0.93240 =$ **1.00000**.

Uncompensated drift of 130 km is over 25% of the route—demonstrating why relative velocity navigation is mandatory.

Part VI: Master Reference Summary & Formula Sheet

Chapter 12: Comprehensive Formula Matrix & Fast-Track Index

This master reference section synthesizes all governing kinematic equations across 1D motion, 2D projectile dynamics, circular kinematics, and Galilean relative motion directly into an integrated reference matrix:

Domain	Governing Equations	Key Assumptions & Scope
1D Kinematics (SUVAT)	• $v = v_0 + at$ • $\Delta x = \frac{1}{2}(v_0 + v)t$ • $\Delta x = v_0 t + \frac{1}{2}at^2$ • $v^2 = v_0^2 + 2a\Delta x$	Constant acceleration only (a = const). Free-fall under gravity: $a = -g$.
Projectile Motion (Flat Ground)	• $v_{0x} = v_0 \cos\theta, \quad v_{0y} = v_0 \sin\theta$ • Flight Time: $T = 2v_0 \sin\theta / g$ • Max Height: $H = v_0^2 \sin^2\theta / (2g)$ • Range: $R = v_0^2 \sin 2\theta / g$ • Trajectory: $y(x) = x \tan\theta - \frac{gx^2}{2v_0^2\cos^2\theta}$	Zero air resistance. Uniform gravitational field g. Complementary angles: $R(\theta) = R(90° - \theta)$. Universal identity: $R = 4H \cot\theta$.
Circular Kinematics	• Arc length: $s = R\theta$, Speed: $v = \omega R$ • Centripetal: $a_c = v^2/R = \omega^2 R = 4\pi^2 R/T^2$ • Tangential: $a_t = dv/dt = \alpha R$ • Total accel: $a = \sqrt{(a_c^2 + a_t^2)}$ • Direction: $\varphi = \arctan(a_c / a_t)$	Uniform: $a_t = 0$, $a = a_c$. Non-uniform: $a_t \neq 0$. Radius of curvature: $\rho = v^2 / a_\perp$.
Relative Motion (Galilean)	• Position: $\mathbf{r}_{P/A} = \mathbf{r}_{P/B} + \mathbf{r}_{B/A}$ • Velocity: $\mathbf{v}_{P/A} = \mathbf{v}_{P/B} + \mathbf{v}_{B/A}$ • Acceleration: $\mathbf{a}_{P/A} = \mathbf{a}_{P/B}$ (inertial frames)	Speeds $v \ll c$ (classical limit). Subscript cancellation: $P/B + B/A = P/A$.

12.1 Master Index of Hand-Calculation Fast-Tracks

Trick ID	Concept & Technique	Core Formula / Shortcut Rule	Target Applications
TRICK K1	"5-Second Drop" Rule	$\Delta y \approx 5t^2$ (m), $v \approx 10t$ (m/s) • Sub 2%	Free fall mental math
TRICK K2	Galileo's Odd Numbers	$\Delta x_1 : \Delta x_2 : \Delta x_3 \ldots = 1 : 3 : 5 : 7 : \ldots$	Equal interval intervals
TRICK K3	Universal Projectile Identity	$R = 4H \cot \theta$ (At 45°: $R_{max} = 4H$)	Range & height ratios
TRICK K4	Speed at Any Elevation	$v(y) = \sqrt{(v_0^2 - 2g\Delta y)}$ • Angle-independent	Impact & catch speeds
TRICK K5	Equal-Acceleration Angle	$a_r = a_t$ when $\theta^* = \frac{1}{2}$ rad ≈ 28.6° (0.0796 rev)	Non-uniform circle tests
TRICK K6	River Crossing Shortcut	$t_{min} = d/v_{bw}$ • Zero drift: $\sin\theta = v_{wg}/v_{bw}$	Relative boat/wind vectors

12.2 Kinematic Problem-Solving Decision Matrix

Scenario / Motion Category	Key Indicators in Statement	Optimal Mathematical Tool	Primary Trap to Bypass
1D Free-Fall / Dropped Body	Released from rest, $v_0 = 0$, no air drag mentioned	Fast-Track K1: $\Delta y \approx 5t^2$ $v^2 = 2g\Delta y$	Setting g negative when choosing downward as positive direction.
2D Elevated Projectile (Cliff)	Horizontal launch or angled launch where landing $\Delta y \neq 0$	Decouple: $x(t) = v_{0x}t$ Quadratic in $y(t)$ for flight time	Using $R = (v_0^2/g)\sin 2\theta$ (strictly invalid for $\Delta y \neq 0$).
Target Interception	Two bodies moving simultaneously; find collision point	Equate position vectors: $\mathbf{r}_1(t) = \mathbf{r}_2(t)$ Solve for single time parameter t	Attempting to equate velocities instead of spatial coordinates.
Circular Speed Limit	Radius given, constant speed or rate of speed change	$a_c = v^2/R = 4\pi^2R/T^2$ If speeding: $a = \sqrt{(a_c^2 + a_t^2)}$	Forgetting centripetal acceleration vector changes direction continuously.
Crosswind / Current Drift	Heading vs track; shortest time vs shortest path	Vector addition: $\mathbf{v}_{P/E} = \mathbf{v}_{P/M} + \mathbf{v}_{M/E}$ Fast-Track K6	Confusing airspeed/boat speed with resultant groundspeed.

Chapter 13: Examination Pitfalls & Systematic Problem-Solving Protocol

⚠ Top 5 Examination Pitfalls & Conceptual Traps

1. **Confusing Zero Vertical Velocity with Zero Acceleration at Apex:** At the apex of projectile motion, $v_y = 0$, but the acceleration is **strictly non-zero** ($a_y = -g = -9.80\ \text{m/s}^2$ downward). If acceleration were zero at the apex, the projectile would stay suspended in mid-air!
2. **Misapplying Flat-Ground Range Formula to Cliff Drops:** The formula $R = v_0^2 \sin 2\theta / g$ holds **only when the landing elevation equals the launch elevation ($\Delta y = 0$)**. For drops from cliffs or elevated targets, one must solve the full quadratic $y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2$.
3. **Assuming Uniform Circular Motion Has Zero Acceleration:** Because the speed $|\mathbf{v}|$ is constant, students frequently forget that velocity is a vector. The changing direction produces a non-zero centripetal acceleration $a_c = v^2/r$ directed toward the center.
4. **Neglecting Downstream Drift in Minimum-Time River Crossing:** To cross a river in the shortest possible time, head directly across ($\perp$ to banks). The boat lands downstream due to river current. Aiming upstream eliminates drift but *increases* the crossing time!
5. **Sign Errors in Galilean Velocity Addition:** Ensure subscripts follow the chain rule: $\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C}$. If given $\mathbf{v}_{C/B}$, you must negate it: $\mathbf{v}_{B/C} = -\mathbf{v}_{C/B}$.

★ The 5-Step Systematic Kinematics Problem-Solving Protocol

1. **Coordinate Frame Selection:** Define a fixed Cartesian origin (typically launch or starting location). Clearly state positive x (forward) and positive y (upward) directions.
2. **Parameter Extraction:** List all known variables with explicit SI units: x_0, y_0, v_{0x}, v_{0y}, a_x, a_y, and identify target unknowns.
3. **Axis Decoupling:** Deconstruct 2D motion into mutually independent horizontal and vertical 1D kinematics. Select the SUVAT equation whose missing variable matches your problem requirements.
4. **Symbolic Isolation:** Solve all algebraic equations symbolically for target quantities *before* substituting numerical figures. Check dimensional consistency.
5. **Physical Sanity Check:** Test boundary cases, verify conservation of energy equivalence ($\Delta v^2 = 2g\Delta y$), and check orders of magnitude using Fast-Track rules.

✓ Standard Physical Constants & SI Unit Notation

• Standard gravitational acceleration: $g = 9.80665\ \text{m/s}^2 \approx 9.80\ \text{m/s}^2$ (standard textbook convention)
• Unit conversions: 1 km/h = (1000 m)/(3600 s) = 1/3.6 m/s ≈ 0.27778 m/s • 1 rev = 2π rad ≈ 6.28318 rad • 1 rad ≈ 57.2958°
• Rigorous SI consistency: Coordinates in meters (m), time in seconds (s), speeds in m/s, acceleration in m/s^2, angles in radians (rad) for calculus and degrees (°) for navigational bearings.